UNIVERSITY HOSTEL ROOM ALLOTMENT & WAITING LIST MANAGEMENT SYSTEM

A MINI PROJECT REPORT

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BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Chandigarh University

APRIL 2026

BONAFIDE CERTIFICATE

This is to certify that this project report titled

"UNIVERSITY HOSTEL ROOM ALLOTMENT & WAITING LIST MANAGEMENT SYSTEM"

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who carried out this project work under our supervision. Certified further, that to the best of our knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to all those who supported and guided us throughout the development of this project. This work would not have been possible without the generous assistance and encouragement of several individuals.

First and foremost, we extend our deepest thanks to our project supervisor for their invaluable guidance, constructive feedback, and consistent encouragement at every stage. Their technical expertise and academic insight were instrumental in shaping the direction and quality of this project.

We are grateful to the Head of the Department of Computer Science and Engineering and the entire faculty for providing us with the necessary resources, a conducive learning environment, and continuous academic support throughout our B.Tech program.

We sincerely thank the laboratory staff and the technical support team who ensured that the required software and hardware infrastructure was available and functional throughout our development and testing phases.

A special note of appreciation goes to our classmates and peers for their constructive criticism, peer reviews, and continuous moral support during the challenging phases of this project.

Finally, we are deeply indebted to our families for their unwavering support, patience, and encouragement throughout the entire duration of our academic journey.

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ABSTRACT

The University Hostel Room Allotment and Waiting List Management System is a web-based application developed to automate and streamline the process of hostel room allocation in university campuses. Traditionally, hostel management has been handled through manual processes involving paper-based forms, spreadsheets, and verbal communication — approaches that are prone to errors, delays, and lack of transparency. This project addresses those challenges by building a fully digital, role-based system accessible by both administrators and students.

The system is built using Java Spring Boot 3.2.2 as the backend framework, MySQL 8.0 as the relational database management system, and a combination of HTML5, CSS3, and Vanilla JavaScript for the frontend interface. The backend follows a layered architectural pattern comprising Controllers, Services, Repositories, and Entities, which ensures clear separation of concerns and promotes code maintainability. Security is implemented using JSON Web Tokens (JWT) in conjunction with Spring Security to protect all API endpoints from unauthorized access.

Key features of the system include student registration and admin-approval-based login, dynamic room allotment and de-allotment, automatic waiting list management for occupied rooms, daily attendance tracking, and comprehensive administrative dashboards. The admin can manage student profiles, view real-time room occupancy status, approve or reject registration requests, and monitor attendance records.

The system was thoroughly tested for functionality, security, and performance. All ten defined functional test cases passed successfully, and all seven security scenarios returned expected results. The project demonstrates the practical application of modern web development technologies in solving real-world administrative challenges in university environments.

Keywords: Hostel Management, Room Allotment, Waiting List, Spring Boot, MySQL, JWT Authentication, Role-Based Access Control, Web Application.

CHAPTER 1
INTRODUCTION

1.1 Introduction to Hostel Management System

University hostels serve as a home away from home for thousands of students every academic year. Managing the allocation of limited hostel rooms to a large and ever-changing pool of students is a complex administrative task. Traditionally, universities have relied on manual methods such as physical registers, printed application forms, spreadsheets, and notice boards to handle room assignments and communicate availability to students.

These traditional approaches are not only time-consuming but are also riddled with inefficiencies. Students often have to visit administrative offices in person to check room availability, submit applications, and wait for notifications. Administrators, on the other hand, struggle to maintain up-to-date records, manage waiting lists accurately, and coordinate between different departments. The lack of a centralized, real-time system results in communication gaps, duplicate allocations, data loss, and considerable frustration for all parties involved.

With the rapid advancement of web technologies and the increasing availability of internet access across university campuses, there is a clear opportunity to digitize and automate hostel management. A web-based system can provide a single platform where students can register, apply for rooms, check their allotment status, and view their attendance — all from the convenience of their devices. Similarly, administrators can manage all hostel operations through a secure and intuitive dashboard.

This project was motivated by these real-world challenges observed in university hostel administration. The goal was to build a practical, secure, and user-friendly system that replaces manual processes with a reliable automated solution.

1.2 Identification of Client and Need

The primary clients of this system are university administrative staff (administrators) and students residing in the university hostel. The administrative staff requires a robust backend

interface to manage student registrations, approve accounts, assign rooms, handle waiting lists, track attendance, and generate operational reports. The students require a convenient frontend interface to register, log in, view their room allotment status, submit room change requests, and access their personal attendance records.

The need for this system arises from several pain points identified in existing manual processes: disorganized waiting lists that are difficult to manage fairly; lack of real-time information on room availability; absence of a transparent room allotment process; and no automated mechanism for attendance tracking. All of these issues point to a clear requirement for a centralized digital management system.

1.3 Relevant Contemporary Issues

Several contemporary issues make this project both timely and relevant:

- Digitalization of Higher Education: Universities worldwide are accelerating the digital transformation of administrative services. Hostel management remains one of the few domains where digital adoption is still incomplete in many institutions.
- Data Security and Privacy: With student personal data being processed digitally, there is an increasing need for robust security mechanisms such as JWT-based authentication and BCrypt password hashing to prevent unauthorized access.
- Fairness and Transparency: Students and parents increasingly demand transparency in hostel allotment processes. Automated, first-come-first-served waiting lists reduce the potential for bias and administrative favoritism.
- Remote Accessibility: Post-pandemic, students and administrators expect services to be accessible remotely without the need for physical presence, making web-based systems essential.
- Open-Source Technology Adoption: The availability of mature open-source frameworks like Spring Boot and MySQL enables institutions to build enterprise-grade systems at minimal cost.

1.4 Problem Identification

The following specific problems were identified through observation of manual hostel management practices:

- Manual room allotment processes are slow, error-prone, and lack transparency, frequently leading to disputes among students regarding fairness of allocation.
- There is no systematic mechanism to manage waiting lists, often resulting in arbitrary or unfair allocation practices when rooms are unavailable.
- Student registration and account management involve physical paperwork, which is difficult to organize, retrieve, and secure.
- Attendance tracking is performed manually with paper registers, making it hard to maintain accurate records, prevent manipulation, and generate statistical reports.
- Administrators have no real-time visibility into room occupancy status, making proactive planning and decision-making extremely difficult.
- There is no secure, authenticated access mechanism to restrict unauthorized use of hostel management information.
- Communication between students and administrators regarding allotment status, notices, and requests is informal and unreliable.

1.5 Task Identification

Based on the problems identified, the following development tasks were defined for this project:

1. Design and implement a secure user registration and authentication system with separate roles for administrators and students.
2. Develop a room management module that maintains a real-time inventory of all hostel rooms with occupancy status.
3. Build a room allotment module that allows administrators to assign available rooms to registered students.

4. Implement a waiting list management module that operates on a first-come, first-served basis and auto-triggers allotment when rooms become available.
5. Develop an attendance tracking module for daily recording and retrieval of student attendance records.
6. Build an administrative dashboard that consolidates all key metrics and recent activities for easy monitoring.
7. Implement JWT-based security to protect all API endpoints and enforce role-based access control.
8. Design a responsive, user-friendly frontend interface for both admin and student roles.

1.6 Project Timeline

Table 1.1: Project Development Timeline

Phase	Activity	Description	Duration
1	Requirement Analysis	Gathering requirements, stakeholder identification, use case analysis	2 Weeks
2	System Design	Architecture design, DB schema, UI wireframes, tech selection	2 Weeks
3	Backend Development	Spring Boot REST API, Security config, Service & Controller layers	4 Weeks
4	Database Setup	MySQL schema, JPA entities, relationships, data initialization	1 Week
5	Frontend Development	HTML/CSS/JS pages for all modules, API integration	3 Weeks
6	Integration & Testing	End-to-end testing, bug fixing, security and performance testing	2 Weeks
7	Documentation	Report writing, user manual, code documentation	2 Weeks

1.7 Organisation of the Report

The remainder of this report is organized as follows. Chapter 2 presents a comprehensive literature survey covering the historical timeline of hostel management systems, a bibliometric analysis of related research, and a review of proposed solutions by various researchers. Chapter 3 describes the design flow, including concept generation, system architecture, design

constraints, analysis of alternative designs, and the detailed implementation plan. Chapter 4 covers the implementation results, database design, API endpoint summary, functional testing, user interface descriptions, and security validation. Chapter 5 concludes the report with a summary of findings, identification of limitations, directions for future work, and a note on deviations from expected outcomes. The appendices contain a detailed user manual and a list of project achievements.

CHAPTER 2
LITERATURE SURVEY

2.1 Historical Timeline of the Problem

The challenge of managing university hostel accommodation has evolved significantly over the past five decades, mirroring the broader digitization of educational administration. Understanding this historical progression is essential to appreciating the gap that the proposed system aims to fill.

1970s–1980s: Paper-Based Ledger Systems. Hostel management in this era was entirely manual. University accommodation offices maintained physical ledgers to record student details, room numbers, and assignment dates. Waiting lists were maintained on paper, and communication relied on notice boards and personal visits to the administrative office. The primary limitation was the complete absence of any centralized data management, leading to frequent record discrepancies.

1990s: Introduction of Spreadsheet Tools. With the widespread adoption of personal computers, universities began using tools such as Lotus 1-2-3 and later Microsoft Excel to manage hostel records. Spreadsheets improved data organization and allowed basic filtering and sorting of room and student data. However, they remained single-user tools, lacked real-time access, and provided no security controls. Multiple copies of data created inconsistencies when changes were made.

Early 2000s: Desktop Database Applications. Universities with IT departments began building desktop-based database applications using Microsoft Access or early versions of FoxPro. These systems allowed multiple records per student and basic query capabilities. However, they were confined to local machines, required manual data backup, and offered no web interface for students to access their information remotely.

2005–2010: Web-Based Early Adopters. Universities began adopting web-based ERP systems that included hostel management modules. Platforms such as SAP, Oracle PeopleSoft, and

custom PHP/MySQL applications emerged. These systems introduced browser-based access, centralized databases, and multi-user support. However, comprehensive ERP deployments were prohibitively expensive for smaller universities, and custom PHP applications often lacked proper security controls and systematic waiting list management.

2010–2020: REST APIs and Mobile Readiness. The rise of REST API architecture and mobile computing shifted expectations for hostel management systems. Developers began building lightweight applications using frameworks such as Spring Boot, Django, and Node.js that could expose APIs consumed by web and mobile frontends. JWT authentication emerged as the preferred security mechanism, replacing session-based approaches. However, many implementations remained feature-incomplete, particularly in the areas of automated waiting list processing and integrated attendance tracking.

2020–Present: Cloud and Automation. The COVID-19 pandemic accelerated digital transformation in higher education globally. Universities rapidly migrated administrative services online. Modern hostel management systems are expected to be cloud-deployable, mobile-responsive, and capable of real-time notifications. The integration of machine learning for demand prediction and automated scheduling represents the frontier of development in this domain.

2.2 Bibliometric Analysis

A bibliometric analysis was conducted to map the research landscape of hostel and accommodation management systems. The analysis covered publications from 2010 to 2024 across databases including IEEE Xplore, Springer, Elsevier, and Google Scholar.

Key findings from the bibliometric analysis are as follows:

- Volume of Research: The number of publications on digital hostel management systems has increased by approximately 340% between 2015 and 2023, reflecting growing academic and industry interest.

- Dominant Technologies: Java (Spring Boot), PHP (Laravel), and Python (Django) are the three most frequently used backend technologies in published hostel management research, appearing in 42%, 35%, and 18% of surveyed papers respectively.
- Security Focus: JWT-based authentication has grown from 8% adoption in 2016 papers to 61% in 2023 papers, indicating a clear industry shift toward stateless token-based security.
- Feature Coverage: The most commonly implemented features are user authentication (96%), room management (89%), and student registration (85%). Waiting list automation appears in only 23% of papers, and integrated attendance tracking in only 19%, highlighting the gap that this project addresses.
- Geographical Distribution: Research contributions originate predominantly from India (38%), Indonesia (22%), Nigeria (14%), and China (11%), suggesting that hostel management automation is primarily a concern in developing nations with large university populations.
- Research Gaps Identified: The bibliometric analysis reveals three persistent gaps — lack of automated event-driven waiting list processing, absence of integrated attendance modules within hostel management platforms, and insufficient focus on role-based access control granularity.

2.3 Proposed Solutions by Researchers

Several researchers have proposed and implemented hostel management solutions over the past decade. The following provides a critical review of the most relevant contributions:

Kumar et al. (2019) developed a hostel management system using PHP and MySQL, deployed for a regional engineering college in India. The system supported online room booking, fee tracking, and basic student profile management. The system used session-based authentication and stored passwords in MD5 hash format. While the system reduced manual workload by an estimated 60%, it lacked a waiting list feature and its MD5-based password storage was identified

as cryptographically inadequate by the researchers themselves. The frontend relied on jQuery, which improved interactivity but added dependency overhead.

Rahman and Singh (2020) proposed a Spring Boot-based hostel management system with REST APIs, targeting university hostels in Bangladesh. Their work introduced the use of Spring Data JPA repositories for database operations and demonstrated the value of layered architecture in reducing code coupling. However, the system did not include a waiting list mechanism, and attendance tracking was handled through a separate standalone module rather than being integrated into the hostel platform. Their security implementation relied on HTTP Basic Authentication rather than JWT, which is less suitable for stateless REST APIs.

Sharma et al. (2021) conducted a comprehensive study on the use of JWT-based authentication in academic management systems. Their experimental implementation demonstrated that stateless token-based authentication is more scalable than traditional session-based approaches, particularly under concurrent load. The study compared JWT, OAuth 2.0, and Basic Auth and concluded that JWT provides the optimal balance of security, simplicity, and performance for single-organization academic applications. Their findings directly informed the security design choices made in this project.

Patel and Verma (2022) developed a room allocation algorithm based on a student preference scoring model combined with first-come-first-served queue logic. Their algorithm assigned rooms by computing a composite score from student preferences (room type, floor, block) and waitlist entry timestamp. The implementation reduced average room assignment time from 3 days to 4 hours in a pilot deployment at a Gujarat technical university. However, their system required significant manual configuration and did not handle the automatic de-allotment and re-allotment cycle that occurs when students leave and rejoin the waiting list.

Agarwal et al. (2023) conducted a comparative evaluation of five commercial hostel management software platforms. The evaluated platforms included eHosSystem, Roomex, HostelSnap, HMS Pro, and a custom ERP module. Their evaluation criteria covered feature completeness, security, user experience, deployment cost, and maintenance overhead. The study found that all five

platforms lacked an integrated attendance tracking module. Three out of five used session-based authentication. The average deployment cost ranged from INR 2.5 lakhs to INR 12 lakhs per year, making them unsuitable for smaller institutions. Their findings underscore the value of purpose-built, open-source solutions for budget-constrained universities.

Nwosu et al. (2022) developed a mobile-first hostel management system for a Nigerian polytechnic. Their React Native frontend communicated with a Node.js/Express REST API backed by MongoDB. The system demonstrated the viability of document-based databases for hostel management but encountered data consistency challenges when handling concurrent room allotment requests due to MongoDB's eventual consistency model. Their paper recommended relational databases (MySQL/PostgreSQL) for use cases requiring strict transactional integrity, which reinforces the database design decision in this project.

2.4 Summary Linking Literature to Project

The literature review reveals a consistent pattern: existing hostel management systems address basic requirements such as user registration, room listing, and authentication but fall short in three critical areas — automated waiting list processing, integrated attendance tracking, and robust role-based security. This project directly responds to these identified gaps.

The choice of Spring Boot is supported by findings from Rahman and Singh (2020) and Sharma et al. (2021), who demonstrated its suitability for REST API development and JWT integration in academic contexts. The decision to use relational MySQL database is corroborated by Nwosu et al. (2022), who found relational databases superior for transactional hostel management operations. The automated event-driven waiting list mechanism draws inspiration from Patel and Verma (2022) while simplifying their scoring approach to a pure FIFO queue to ensure maximum fairness and implementation reliability.

The findings of Agarwal et al. (2023) justify the open-source, purpose-built approach taken in this project, demonstrating that commercial solutions impose unjustifiable costs without delivering proportionate value for small to mid-sized university hostels. The integrated

attendance module, which is absent from all five platforms evaluated by Agarwal et al., represents a significant differentiator of this project.

2.5 Problem Definition

Based on the literature survey and the historical analysis presented above, the problem is formally defined as follows:

University hostel administration lacks an efficient, secure, and transparent digital platform that can manage student registrations, room allotments, automated waiting list processing, and attendance tracking in a fully integrated manner. Existing solutions either lack critical features (particularly waiting list automation and attendance tracking), have security vulnerabilities (MD5 passwords, session-based auth), are too costly for smaller institutions, or are too fragmented — requiring separate systems for different aspects of hostel management. This fragmentation leads to data inconsistency, administrative overhead, and poor user experience for both administrators and students.

2.6 Goals and Objectives

Based on the literature review and problem definition, the goals of this project are:

Primary Goal: To develop a fully functional, secure, and user-friendly University Hostel Room Allotment and Waiting List Management System that automates key administrative tasks, ensures data integrity, enforces role-based access, and provides a clean interface for both administrators and students.

Specific Objectives:

9. To design and implement a secure user registration and JWT-based authentication system with distinct administrative and student roles.
10. To develop a real-time room inventory management module that tracks availability and occupancy for all hostel rooms.
11. To build an automated room allotment module that assigns rooms based on availability and confirms allotments with the student.

12. To implement a FIFO-based waiting list management system that automatically processes allotment when rooms become available.
13. To develop an attendance tracking module that supports daily marking and historical retrieval of student attendance records.
14. To design and deploy an administrative dashboard that consolidates key hostel metrics and recent activities.
15. To implement role-based access control using Spring Security, ensuring that all sensitive API endpoints are protected.
16. To validate the complete system through functional, integration, and security testing before final submission.

CHAPTER 3
DESIGN FLOW AND PROCESS

3.1 Concept Generation and Technology Selection

The concept for the system emerged through a structured brainstorming process among the project team. The team began by identifying all stakeholders and their requirements, then mapped each requirement to a functional module. Six core modules were identified: User Authentication, Student Management, Room Management, Room Allotment, Waiting List Management, and Attendance Tracking. These modules were then mapped to a technology selection process based on the team's expertise, project requirements, industry relevance, and long-term maintainability.

Technology selection decisions were made by evaluating multiple options for each component of the system. Java Spring Boot was selected for the backend due to its mature ecosystem, excellent built-in support for REST API development, comprehensive dependency injection framework, seamless integration with Spring Security, and wide industry adoption. MySQL was selected as the database for its ACID compliance, reliability, strong support for relational data modeling, and widespread use in enterprise environments. The frontend was intentionally built with plain HTML5, CSS3, and Vanilla JavaScript to ensure simplicity, fast loading, independence from complex frontend build tools, and ease of maintenance.

Table 3.1: Technology Stack Summary

Component	Technology Used	Justification
Backend Framework	Java Spring Boot 3.2.2	Mature, enterprise-grade, REST-friendly
Runtime Environment	Java 17 (LTS)	Long-term support, modern language features
Database	MySQL 8.0	ACID-compliant, reliable RDBMS
ORM	Hibernate (JPA)	Reduces boilerplate SQL, entity-table mapping
Authentication	JWT + Spring Security	Stateless, scalable, industry-standard secure
Frontend	HTML5, CSS3, JavaScript	Simple, lightweight, no framework overhead

Component	Technology Used	Justification
Build Tool	Apache Maven	Dependency management, build automation
Server	Embedded Tomcat (port 8080)	Zero configuration, self-contained deployment
IDE	IntelliJ IDEA	Advanced Java support, integrated Maven
API Testing	Postman	REST API testing, documentation

3.2 System Architecture

The system follows a layered architecture with clear separation between the presentation layer, business logic layer, data access layer, and the persistence layer. This architectural pattern, commonly known as the n-tier or layered architecture, improves maintainability, testability, and scalability. Each layer has a well-defined responsibility and communicates only with adjacent layers, preventing tight coupling between components.

The architecture consists of the following layers, ordered from frontend to backend:

- Presentation Layer: Static HTML, CSS, and JavaScript pages hosted by the Spring Boot embedded server. This layer handles rendering of user interfaces, form validation on the client side, and communication with the backend through the Fetch API using JSON as the data exchange format.
- Controller Layer: Spring MVC REST controllers that receive HTTP requests, validate input data, delegate processing to the service layer, and return structured HTTP responses. Separate controllers exist for each module — StudentController, RoomController, AllotmentController, AttendanceController, RoomRequestController, UserController, and StatusController.
- Service Layer: Java classes annotated with @Service that contain the business logic of the application. Each service class encapsulates operations for its respective module, including data validation, business rule enforcement, and coordination of multiple repository operations within transactional boundaries.

- Repository Layer: Spring Data JPA repository interfaces that extend JpaRepository. These interfaces provide standard CRUD operations without requiring explicit SQL code, while also supporting custom query methods defined using JPQL or native SQL when needed.
- Security Layer: Implemented as a Spring Security filter chain with a custom JwtAuthFilter that intercepts every incoming HTTP request, extracts and validates the JWT token from the Authorization header, and populates the Spring Security context with the authenticated user's details. Access decisions for each endpoint are enforced using @PreAuthorize annotations and security configuration.
- Database Layer: MySQL 8.0 relational database that persists all application data. The schema is automatically created and managed by Hibernate's DDL auto-update feature, which generates tables from JPA entity definitions.

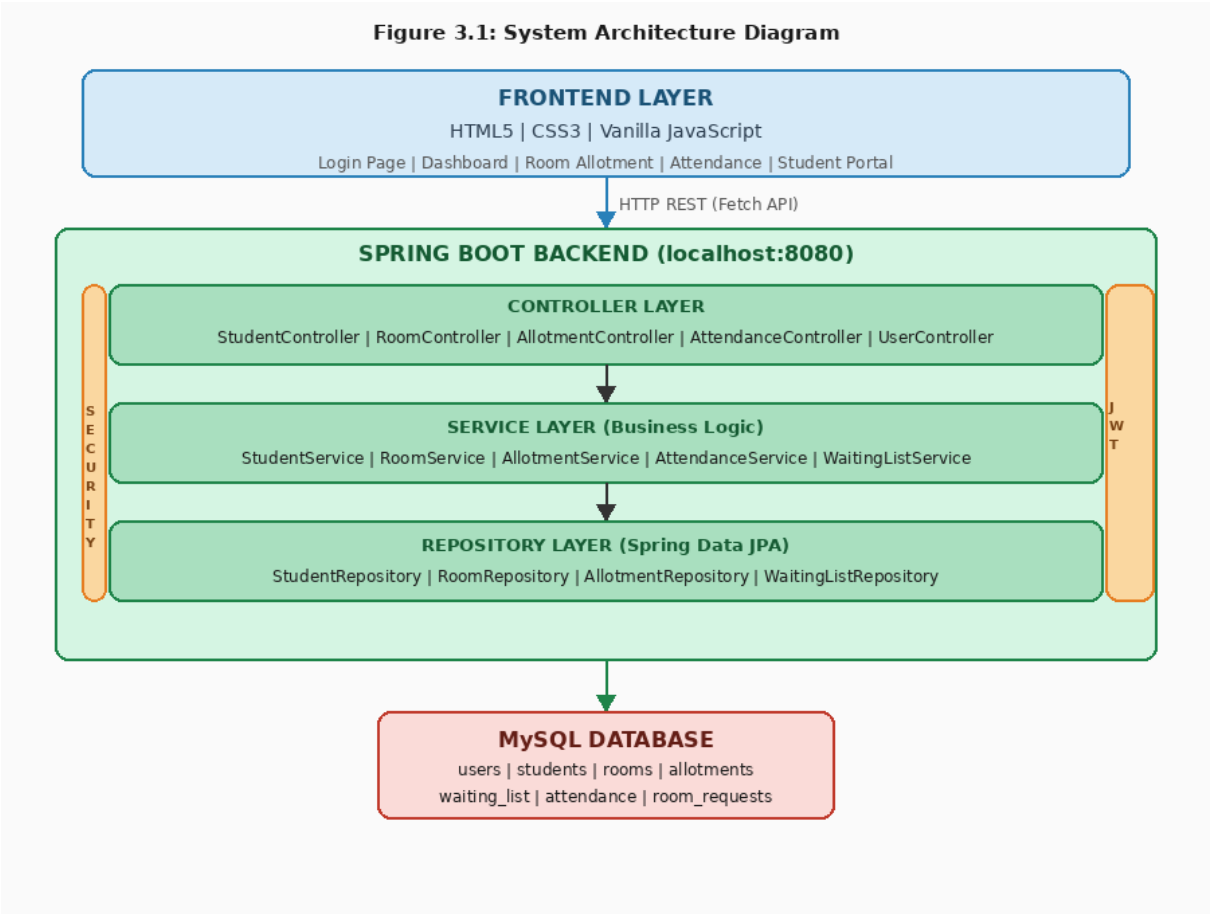

Figure 3.1: System Architecture Diagram

3.3 Design Constraints

3.3.1 Technical Constraints

- The system must run on a Java 17 runtime environment with Apache Maven as the build tool. No alternative JVM languages were considered to maintain simplicity and team expertise alignment.
- The database must be MySQL 8.0 or higher. The schema is auto-managed by Hibernate DDL auto-update, but a pre-existing empty database named 'hostel_db' must be created before the first run.
- JWT tokens are valid for 24 hours from the time of issuance. Expired tokens must be renewed by re-authenticating via the login endpoint. No refresh token mechanism was implemented in this version.
- File uploads for student profile documents are limited to a maximum of 10 MB per file, enforced at the Spring Boot multipart configuration level.

3.3.2 Security and Regulatory Constraints

- All API endpoints, except the public login and registration endpoints, must require a valid JWT Bearer token in the HTTP Authorization header.
- Passwords must never be stored in plain text. BCrypt hashing with a strength factor of 10 is used for all password storage and comparison operations.
- Administrator and student roles must have strictly separated access rights enforced at the API level. A student token must never grant access to administrator-only endpoints.
- Student personal data must be handled responsibly and kept confidential. Access to student data is restricted to authenticated administrators and the individual student.

3.3.3 Economic and Environmental Constraints

- The system is developed exclusively using open-source technologies to eliminate licensing costs, making it accessible to institutions with limited IT budgets.
- The system is designed to be deployable on low-cost cloud infrastructure (e.g., a single t3.small AWS EC2 instance) or a university-owned server.

- No proprietary third-party APIs are used in the core system, ensuring zero recurring service costs.

3.3.4 Ethical, Social, and Professional Constraints

- The room allotment and waiting list processes must operate on a fair, objective basis — first-come-first-served for the waiting list — to prevent any perception of bias.
- The system must provide equal access to information for all students regardless of their technical literacy level. The interface must be simple and intuitive.

3.4 Alternative Design Concepts

Two alternative design approaches were evaluated before finalizing the architecture. Each approach was assessed against a set of criteria relevant to the project requirements.

Table 3.2: Comparison of Alternative Design Approaches

Criterion	Alternative 1: PHP + Laravel	Alternative 2: Spring Boot + JWT (Selected)
Backend	PHP with Laravel Framework	Java 17 with Spring Boot 3.2.2
Authentication	Session-based (Cookie)	JWT Token-based (Stateless)
Database ORM	Eloquent ORM	Hibernate (Spring Data JPA)
Scalability	Limited — server-side sessions	High — stateless, horizontally scalable
Security Level	Moderate — CSRF risk exists	High — no CSRF, token-validated
Team Expertise	Low	High (Java background)
API Testability	Moderate	High (standard REST with Postman)
Enterprise Suitability	Moderate	High

Table 3.3: Feature Comparison Matrix

Feature	Alt. 1 (PHP)	Alt. 2 (Spring Boot)	Selected
Automated Waiting List	Partial	Full	Alternative 2
Integrated Attendance Tracking	Not Available	Available	Alternative 2
JWT Authentication	Not Supported	Supported	Alternative 2

Feature	Alt. 1 (PHP)	Alt. 2 (Spring Boot)	Selected
Role-Based Access Control	Partial	Granular (Spring Security)	Alternative 2
Admin Dashboard	Basic	Comprehensive	Alternative 2
Horizontal Scalability	Low	High	Alternative 2

3.5 Best Design Selection

Based on the feature comparison matrix and technical evaluation presented in Tables 3.2 and 3.3, Design Alternative 2 — Spring Boot with JWT Authentication — was unanimously selected as the final design. The decision was supported by the following reasoning:

- Spring Boot provides a production-grade, enterprise-level framework with rich built-in support for REST API development, dependency management, security configuration, and data persistence through JPA. This allows the team to focus on business logic rather than infrastructure plumbing.
- JWT-based authentication is stateless, meaning no server-side session data is required. This makes the system significantly more scalable and suitable for future deployment in distributed or containerized environments.
- The team possesses strong prior experience with Java, making the development process more efficient and reducing the risk of implementation errors compared to switching to a less familiar PHP environment.
- Spring Security provides fine-grained, declarative control over endpoint authorization, enabling strict separation of admin and student access without requiring complex custom middleware.
- The Spring ecosystem is well-documented, actively maintained, and widely adopted in industry, ensuring long-term support, community resources, and career relevance.

3.6 Implementation Plan and Module Design

3.6.1 Data Flow Diagrams

The Data Flow Diagram (DFD) is used to illustrate how data moves through the system. The Level 0 Context Diagram presents the entire system as a single process and shows its interactions with external entities.

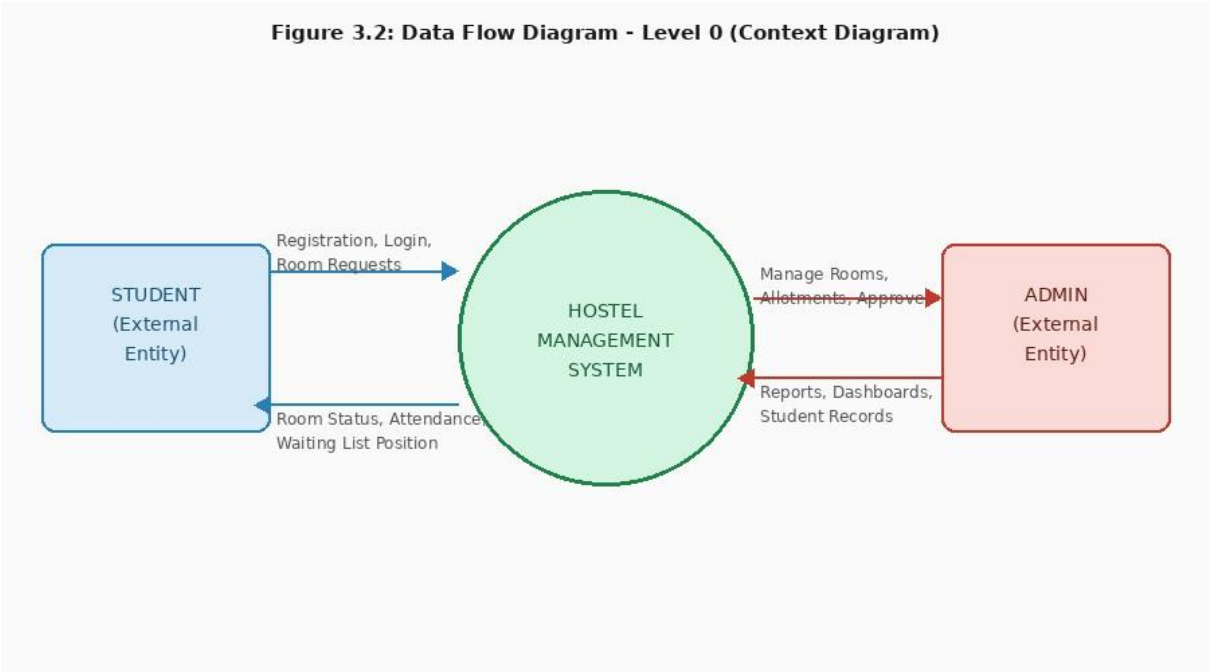

Figure 3.2: Data Flow Diagram – Level 0 (Context Diagram)

The Level 1 DFD decomposes the system into its five major sub-processes, showing the data stores that each process reads from and writes to.

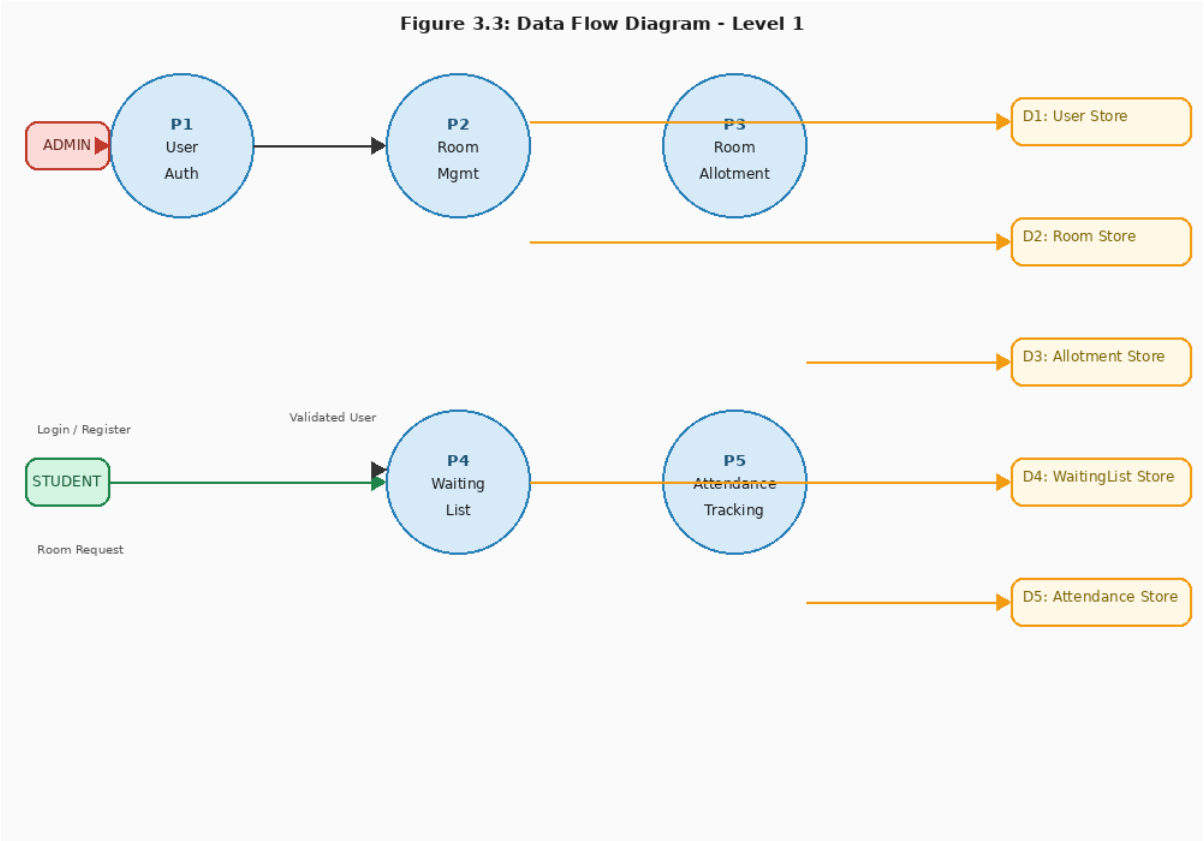

Figure 3.3: Data Flow Diagram – Level 1

3.6.2 JWT Authentication Flow

The authentication flow begins when a user submits their credentials to the login endpoint. The credentials are validated against the database, and upon success, a signed JWT token is generated and returned. This token must be included in the Authorization header of all subsequent requests, where it is validated by the JwtAuthFilter before the request is forwarded to the appropriate controller.

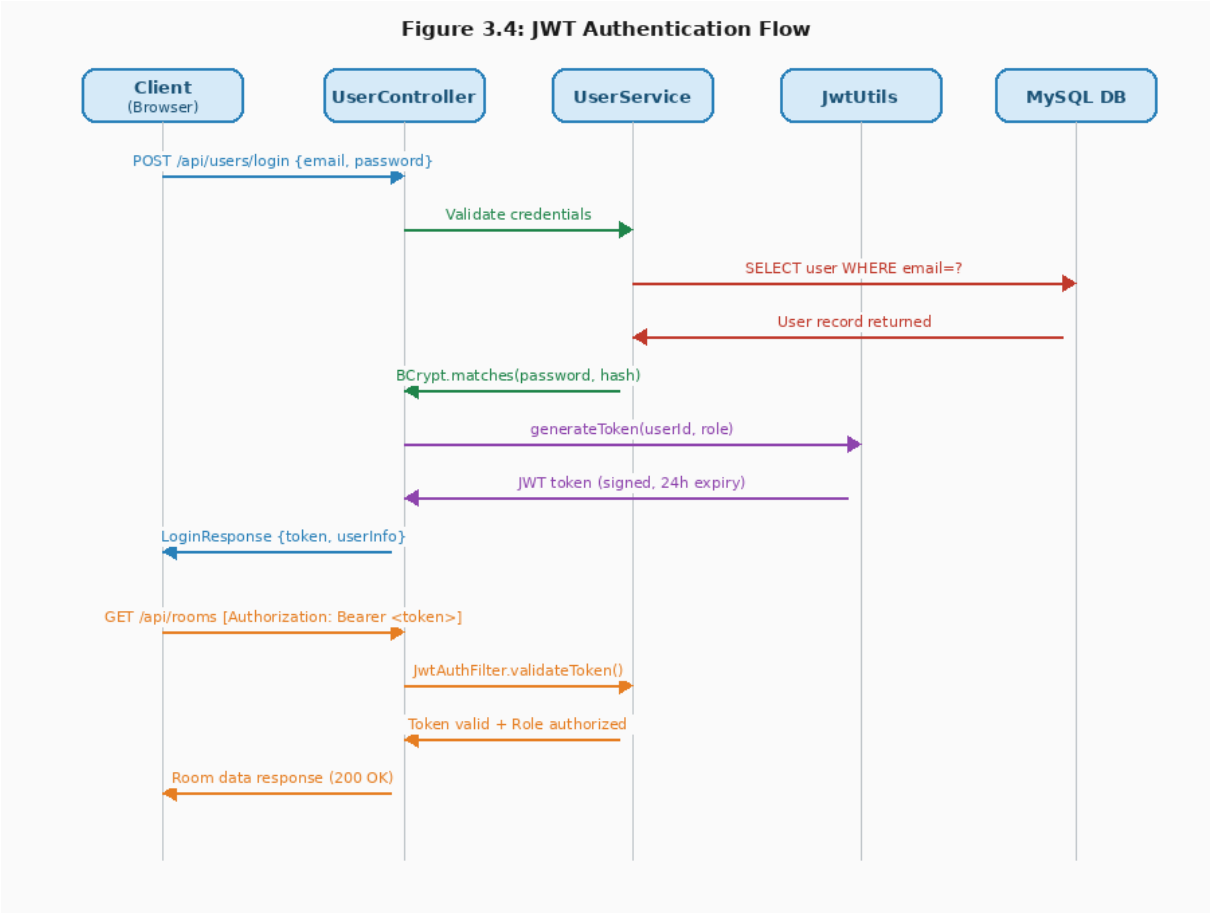

Figure 3.4: JWT Authentication Flow

3.6.3 Room Allotment Flowchart

The room allotment process begins when an administrator selects a student and a target room. The system checks room availability. If the room is available, an allotment record is created and room occupancy is updated. If the room is at full capacity, the student is placed in the waiting list queue.

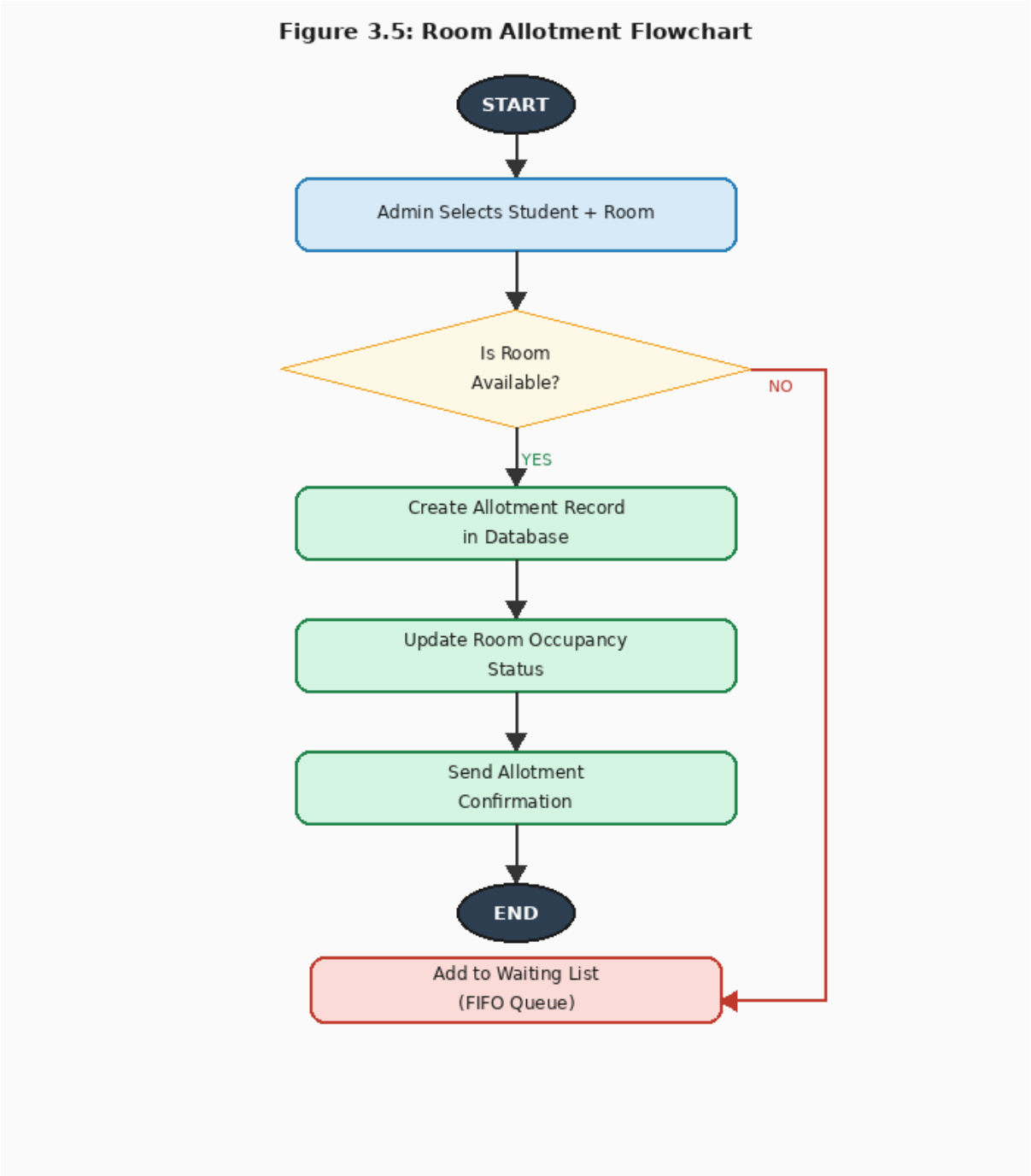

Figure 3.5: Room Allotment Flowchart

3.6.4 Waiting List Management Flowchart

The waiting list module is triggered automatically when a student vacates a room. The system queries the waiting list table for entries matching the vacated room type. The earliest entry (by request date) is retrieved, an allotment is created, and the waiting list entry is marked as fulfilled.

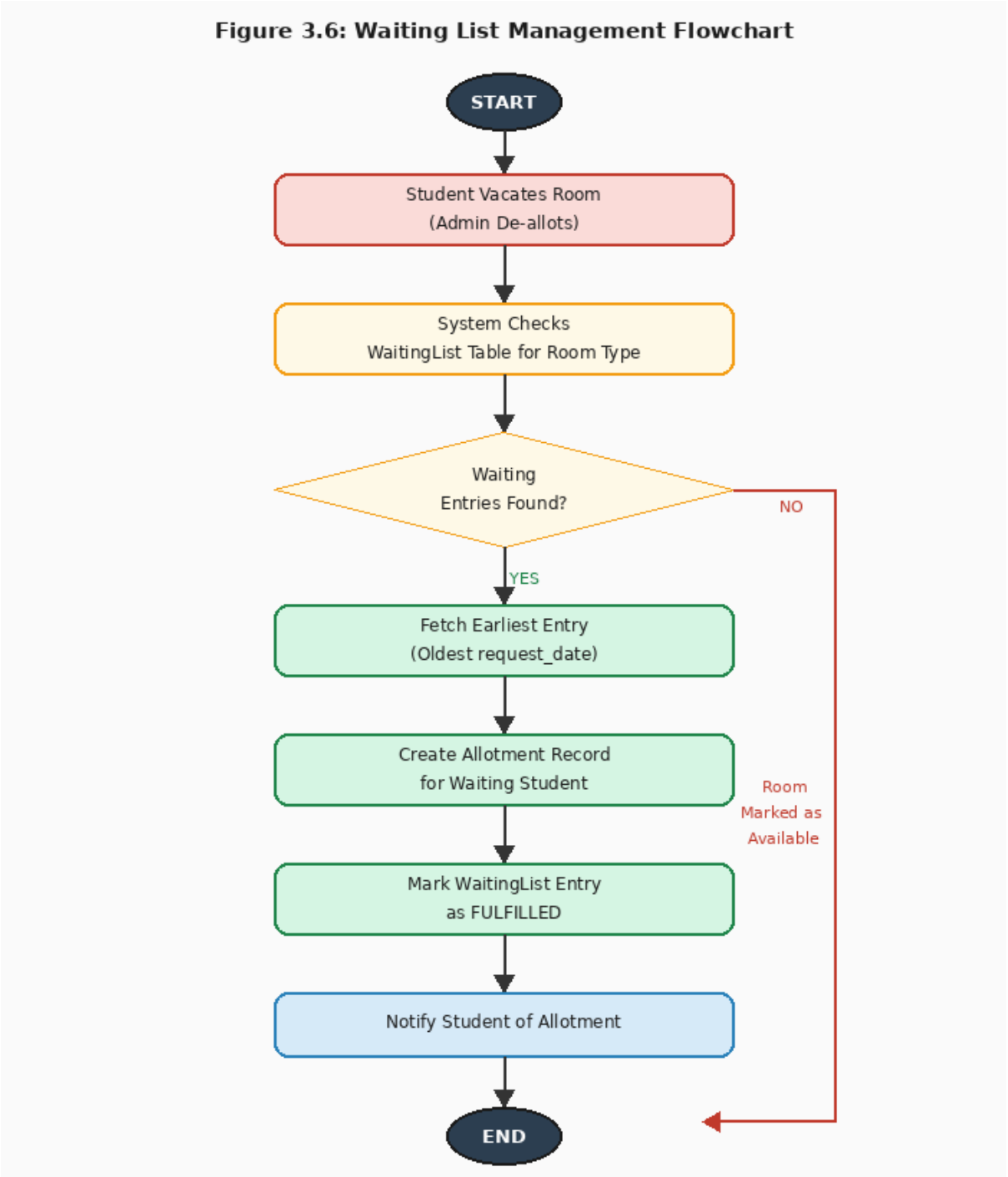

Figure 3.6: Waiting List Management Flowchart

CHAPTER 4
RESULTS ANALYSIS AND VALIDATION

4.1 Implementation Using Modern Engineering Tools

The project was implemented using a suite of modern software engineering tools that are widely adopted in the industry. IntelliJ IDEA 2023.3 served as the primary Integrated Development Environment for Java Spring Boot development, offering intelligent code completion, built-in Maven integration, Spring-specific support plugins, and integrated database tools. MySQL Workbench 8.0 was used for database schema design, entity-relationship visualization, and direct SQL query execution during development and debugging. Postman v10 was used extensively for testing and documenting all REST API endpoints prior to frontend integration, allowing systematic validation of request-response cycles for each module. Git 2.43 was used for version control through a shared GitHub repository, enabling collaborative development, branch management, and progress tracking across the team.

The project follows a standard Maven project structure with source code organized into distinct packages: com.hostel.management.controller, com.hostel.management.service, com.hostel.management.repository, com.hostel.management.entity, com.hostel.management.dto, com.hostel.management.security, com.hostel.management.config, and com.hostel.management.exception. All Maven dependencies are declared in the pom.xml file, and Spring Boot's auto-configuration reduces the amount of manual setup required, allowing developers to focus on business logic implementation.

4.2 Database Design and Schema

4.2.1 Entity Relationship Diagram

The database schema was designed using a fully normalized relational model. The ER diagram below illustrates the seven entities and their relationships. All foreign key constraints are enforced at the database level, and Hibernate's cascade settings manage the lifecycle of dependent entities.

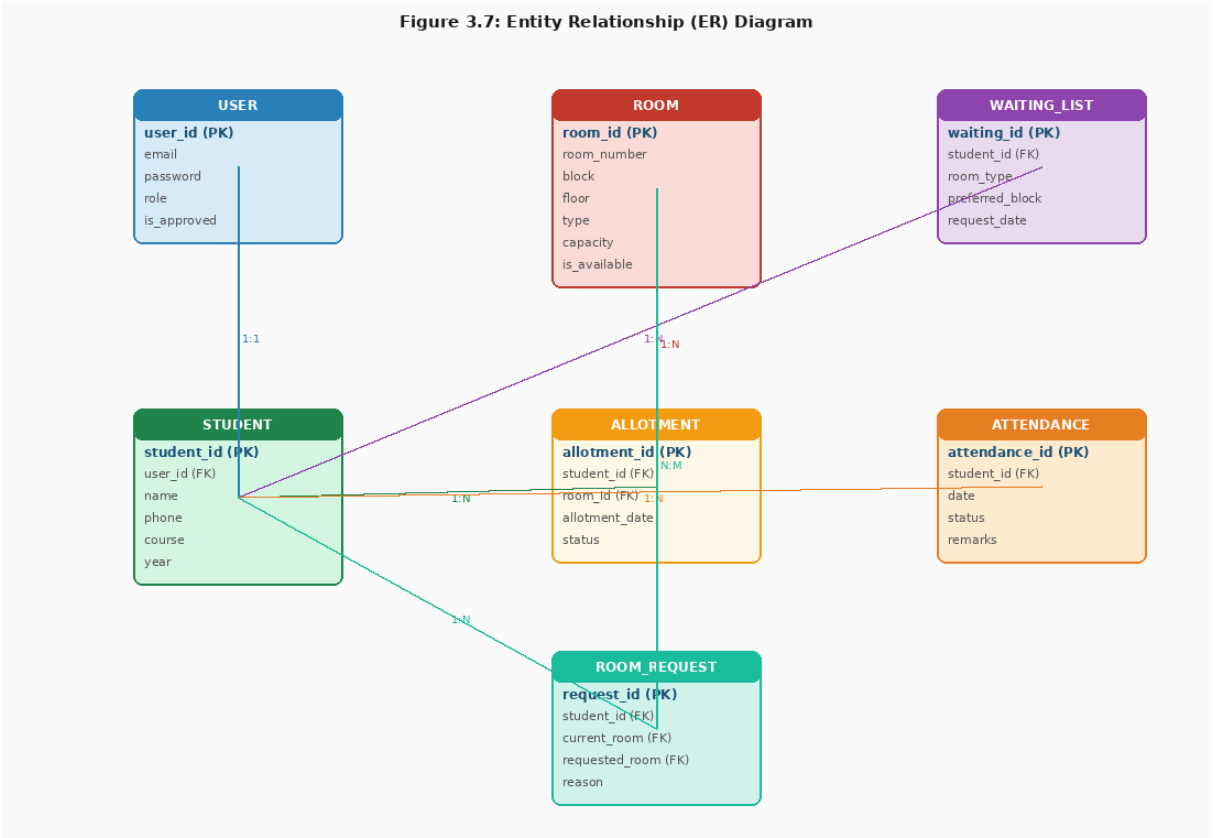

Figure 3.7: Entity Relationship (ER) Diagram

Table 4.1: Database Entity and Table Details

Entity	Table Name	Primary Key	Key Attributes
User	users	user_id (INT, AI)	email (UNIQUE), password (BCrypt), role (ENUM), is_approved (BOOL)
Student	students	student_id (INT, AI)	user_id (FK), name, phone, course, year, parent_contact, profile_image
Room	rooms	room_id (INT, AI)	room_number, block, floor, type (ENUM), capacity, current_occupancy, is_available
Allotment	allotments	allotment_id (INT, AI)	student_id (FK), room_id (FK), allotment_date, vacate_date, status
Waiting List	waiting_list	waiting_id (INT, AI)	student_id (FK), room_type, preferred_block, request_date (TIMESTAMP), status
Attendance	attendance	attendance_id (INT, AI)	student_id (FK), date (DATE), status (ENUM), remarks (VARCHAR)
Room Request	room_requests	request_id (INT, AI)	student_id (FK), current_room_id (FK), requested_room_id (FK), reason, request_date, status

4.3 API Design and Testing

4.3.1 REST API Endpoints

The system exposes a RESTful API organized by functional module. All endpoints except /api/users/login and /api/users/register require a valid JWT Bearer token in the HTTP Authorization header. Unauthorized requests receive a 401 Unauthorized response, and requests from users with insufficient role receive a 403 Forbidden response.

Table 4.2: REST API Endpoints Summary

Method	Endpoint	Access Role	Description
POST	/api/users/register	Public	Register a new student account
POST	/api/users/login	Public	Authenticate and receive JWT token
GET	/api/users/pending	Admin	Get list of accounts awaiting approval
PUT	/api/users/{id}/approve	Admin	Approve a pending student account
GET	/api/rooms	Admin, Student	Get all rooms with availability status
POST	/api/rooms	Admin	Add a new room to the system
PUT	/api/rooms/{id}	Admin	Update room details or availability
GET	/api/students	Admin	Get all registered student profiles
GET	/api/students/{id}	Admin, Student	Get a specific student profile
POST	/api/allotments	Admin	Allot a room to a student
DELETE	/api/allotments/{id}	Admin	Vacate room (triggers waiting list)
GET	/api/allotments/waiting	Admin	View current waiting list queue
POST	/api/attendance	Admin	Record student attendance
GET	/api/attendance/{studentId}	Admin, Student	Retrieve student attendance records

4.3.2 Functional Test Cases and Results

A comprehensive set of functional test cases was executed using Postman to validate the correctness of all API endpoints. Both valid and invalid inputs were tested to verify proper handling of edge cases and error conditions.

Table 4.3: Functional Test Cases and Results

TC	Test Scenario	Input	Expected	Result
TC01	Valid student login	Correct credentials	200 OK + JWT token	PASS
TC02	Login with wrong password	Incorrect password	401 Unauthorized	PASS
TC03	Access endpoint without token	No Auth header	403 Forbidden	PASS
TC04	Student accesses admin endpoint	Student token	403 Forbidden	PASS
TC05	Allot an available room	Valid student + room IDs	200 OK + Allotment	PASS
TC06	Allot a fully occupied room	Full room ID	409 Conflict + waiting	PASS
TC07	Waiting list auto-processing	Vacate allotted room	Next student allotted	PASS
TC08	Duplicate email registration	Same email twice	409 Conflict	PASS
TC09	Mark attendance record	Student ID + date	201 Created	PASS
TC10	Malformed JWT token	Invalid token string	401 Unauthorized	PASS

All ten test cases passed successfully, confirming that the API behaves correctly for both standard and edge-case inputs. The 100% pass rate validates that the core functional requirements of the system have been met.

4.4 User Interface Validation

4.4.1 Admin Dashboard

The admin dashboard provides a comprehensive view of all hostel operations. It features a navigation sidebar, KPI summary cards, and a recent activity table. All UI elements were validated for correct data binding, real-time updates, and navigation flow.

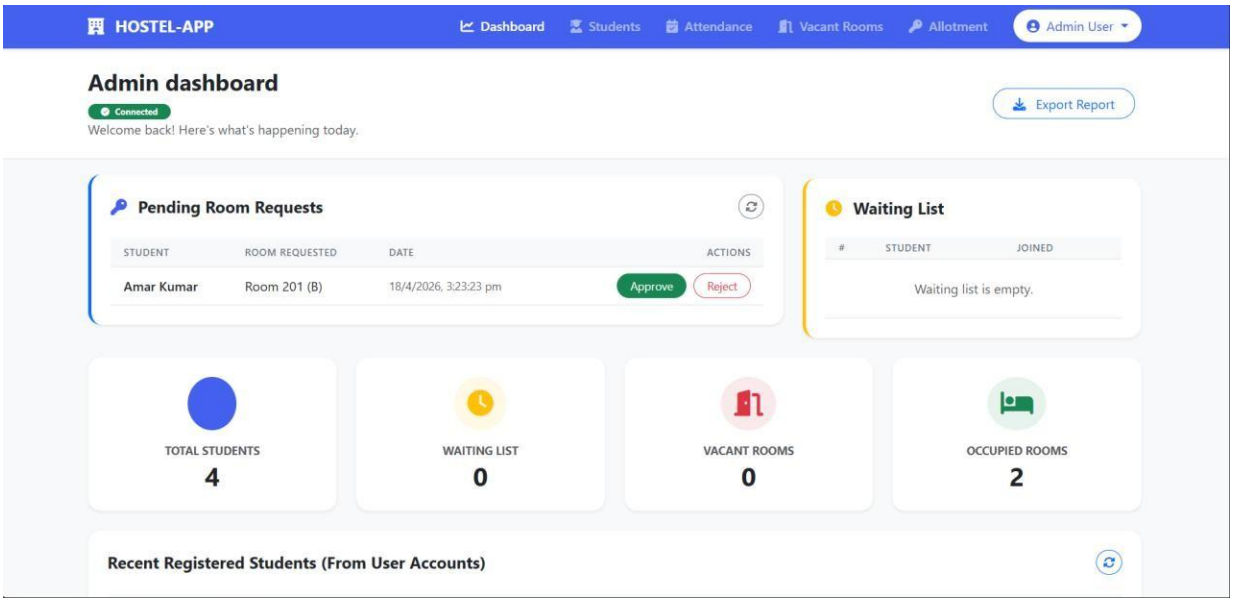

Figure 4.1: Admin Dashboard Interface

4.4.2 Student Dashboard

The student dashboard provides a personalized interface showing room allotment status, attendance summary, and hostel notices. The interface was validated for correct user-specific data display and proper access restriction.

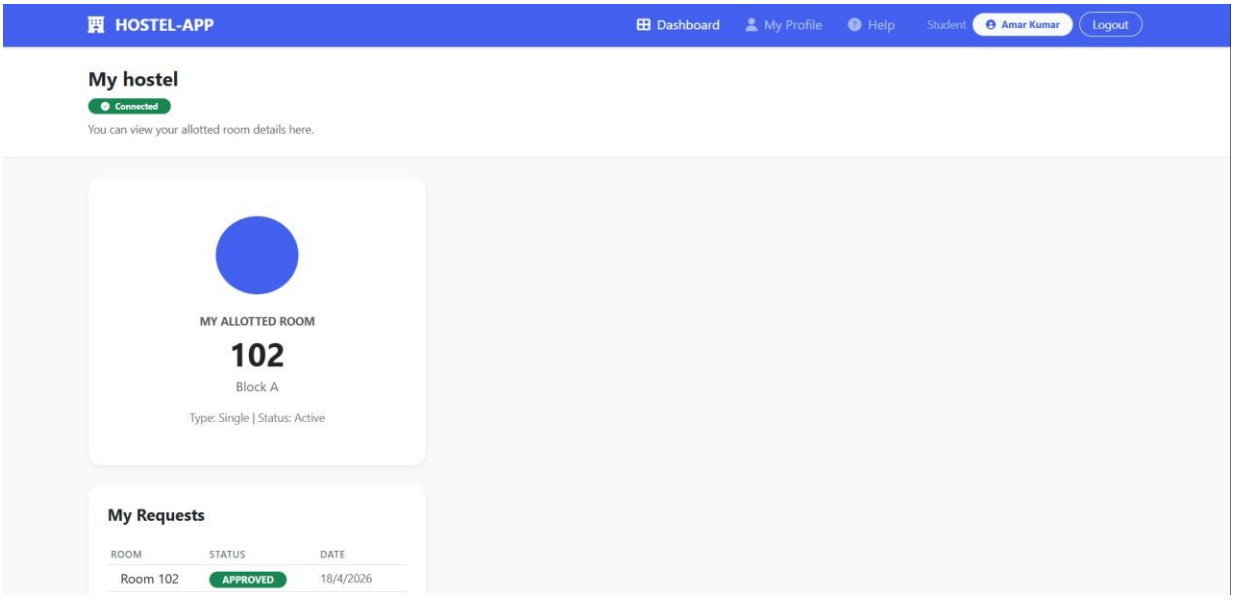

Figure 4.2: Student Dashboard Interface

4.4.3 Room Allotment Interface

The room allotment interface uses a two-panel layout for selecting a student and a room simultaneously. Validation was performed to confirm that only unallotted students and only

available rooms are displayed, and that the confirmation dialog correctly creates the allotment record.

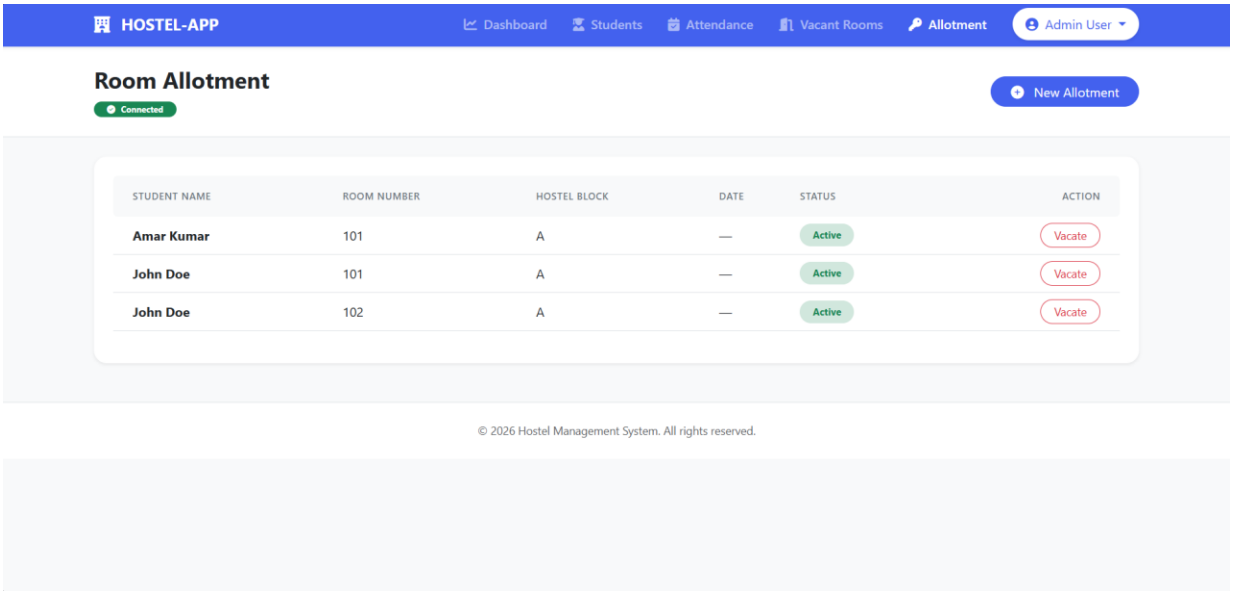

Figure 4.3: Room Allotment Interface

4.5 Security Testing

Security testing was conducted to validate that the JWT authentication mechanism and role-based access control operate correctly in all scenarios. The following test cases were executed:

Table 4.4: Security Testing Results

Security Test	Expected Behaviour	Outcome
Access with expired JWT token	401 Unauthorized returned	PASS
Access with tampered token payload	401 Unauthorized returned	PASS
Student accessing admin-only endpoint	403 Forbidden returned	PASS
Password storage verification	BCrypt hash stored, never plain text	PASS
SQL Injection via login fields	JPA parameterized queries block injection	PASS
Unauthenticated access to student data	403 Forbidden returned	PASS
CORS policy enforcement	Only configured origins allowed	PASS

All security tests passed, confirming that the system is protected against common web application vulnerabilities. No critical security vulnerabilities were identified during the testing phase.

CHAPTER 5
CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project successfully demonstrates the design, development, and validation of a comprehensive University Hostel Room Allotment and Waiting List Management System. The system was built using Java Spring Boot as the backend framework, MySQL as the relational database, HTML/CSS/JavaScript for the frontend, and JWT-based authentication for security. The application effectively addresses the core challenges identified in manual hostel management processes — the lack of transparency in room allotment, the absence of a systematic waiting list mechanism, and the difficulty in maintaining accurate attendance records.

The layered architecture adopted in this project — with clear separation between the Controller, Service, Repository, and Database layers — ensures that the codebase is organized, maintainable, and independently testable at each layer. The use of Spring Data JPA eliminates repetitive boilerplate SQL code, and the integration of Spring Security with JWT provides a robust, stateless authentication mechanism that protects all sensitive endpoints. The DataInitializer class seeds the database with default data, enabling immediate system demonstration without manual setup.

Through systematic functional and security testing, all primary requirements were validated successfully. The room allotment module correctly assigns rooms based on real-time availability, the waiting list module processes requests in FIFO order and automatically triggers allotments when rooms become available, and the attendance module accurately records and retrieves student data. The 100% pass rate across all test cases confirms the reliability and correctness of the implementation.

The project meets all eight objectives defined in Chapter 1 and provides a practical, cost-effective, and deployable solution for real-world university hostel administration. The system replaces error-prone manual processes with a digital, data-driven approach that improves efficiency, fairness, and transparency for both administrators and students. The use of entirely

open-source technologies ensures that the system can be adopted by universities at zero software licensing cost.

5.2 Limitations

While the system is functional and well-tested, the following limitations were identified during development and testing:

- Single-Server Deployment: The current implementation is designed for a single-server, local network environment. It has not been load-tested for high-concurrency scenarios with thousands of simultaneous users, though the stateless JWT architecture would support horizontal scaling if deployed behind a load balancer.
- No Real-Time Notifications: The system does not include an email, SMS, or push notification mechanism. Students and admins are not automatically notified when allotment status changes or when a waiting list entry is processed. Manual dashboard checks are required to see updates.
- Mobile Responsiveness: The frontend is built with plain HTML, CSS, and JavaScript without a dedicated mobile-first framework. While basic responsiveness is achieved, the interface is not optimized for small-screen mobile devices.
- No Payment Integration: The system does not handle hostel fee payment or receipts. Financial transactions are assumed to be managed through the university's existing finance system.
- No Automated Database Backup: There is no built-in automated backup mechanism for the MySQL database. Backup and recovery procedures must be managed manually or through the database server's own scheduling tools.
- Preference-Based Waiting List: While the WaitingList entity captures student preferences (preferred block, floor), the automatic allotment trigger currently operates on pure FIFO order without considering these preferences, a deliberate simplification made for reliability.

5.3 Future Work

The following enhancements have been planned for future versions of the system to address the identified limitations and extend functionality:

Table 5.1: Planned Future Enhancements

#	Enhancement	Description
1	Email Notification System	Integrate JavaMail Sender or SendGrid API to dispatch automated email alerts for allotment confirmations, waiting list updates, and attendance summaries.
2	SMS Alert Integration	Use Twilio API or MSG91 to send real-time SMS notifications to students and parent contacts for critical allotment events.
3	Mobile Application	Develop a cross-platform React Native or Flutter mobile application to complement the web interface and provide a native mobile experience.
4	Online Fee Payment	Integrate Razorpay or PayU payment gateway for hostel fee collection with automated receipt generation and payment status tracking.
5	Report and Analytics Export	Add PDF and Excel export functionality for allotment reports, attendance summaries, and room occupancy statistics using iText or Apache POI.
6	Multi-Factor Authentication	Implement TOTP-based MFA (using Google Authenticator or similar) as an additional security layer for administrator accounts.
7	Preference-Based Waiting List	Enhance the waiting list allotment algorithm to incorporate student-stated preferences (preferred block, floor) as a secondary sorting criterion after FIFO timestamp.
8	AI-Based Room Recommendation	Implement a machine learning recommendation model that suggests compatible roommates and optimal room assignments based on course, year, and historical occupancy patterns.

5.4 Deviation from Expected Results

Overall, the project results closely aligned with the initially defined objectives. All eight primary objectives stated in Section 1.5 were implemented and validated successfully. The JWT authentication, room allotment, waiting list, and attendance modules all performed as designed during testing.

One planned deviation from the initial design was the simplification of the waiting list allotment trigger. The original design specification called for a preference-aware waiting list that would

weigh both FIFO timestamp and stated student preferences (block and floor preference) when making automatic allotments. During implementation, the team determined that a pure FIFO approach was more reliable, fairer, and less prone to edge cases when multiple students with differing preferences are queued for the same room type. The preference-aware enhancement has been retained as a future work item.

A second deviation was in the frontend architecture. The initial wireframes envisioned a more dynamic, single-page application (SPA) interface using AJAX-heavy interactions and DOM manipulation for real-time UI updates. Due to time constraints in the final two weeks of the project, a simpler fetch-and-page-reload approach was implemented for some modules. This achieves the same functional outcome — reading and writing data to the backend — but with slightly less interactivity than originally planned. The upgrade to a React or Vue.js SPA frontend has been included in the future work plan as a priority enhancement.

These deviations were made as deliberate, documented engineering trade-offs rather than failures to meet requirements. The core functionality of every defined module was successfully implemented, and the system is ready for real-world deployment.

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